6060222

Test Booklet No.

ENGLISH

Test Booklet Code

07

NEET RANKERS TEST SERIES-2025

Do not open this Test Booklet until you are asked to do so.

Important Instructions:

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below:
   (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 and 185). All questions are compulsory.
   (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150, and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.
   Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated**.
3. Each question carries 4 marks. For each correct response, the candidate will get **4 marks**. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720**.
4. **Use Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. **The CODE for this Booklet is 07. Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet.** In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
8. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
9. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
10. Each candidate must show on-demand his/her Admit Card to the Invigilator
11. No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
12. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
13. Use of Electronic/Manual Calculator is prohibited.
14. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
15. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
16. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
17. Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of Scribe or not.

Name of the Candidate (in Capitals): ____________________

Roll Number : In figures ____________________

: In words ____________________

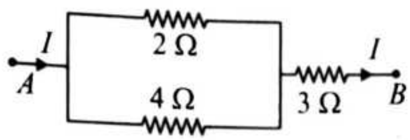
Centre of Examination (in Capitals): ____________________

Candidate's Signature: ____________ **Invigilator's Signature:** ____________

Facsimile Signature stamp of Centre Superintendent

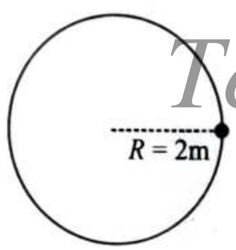
Physics : Section-A (Q. No. 1 to 35)

1. A part of a circuit is shown in the figure. The ratio of heat developed across 2 Ω, 4 Ω and 3 Ω resistances respectively after time t is:



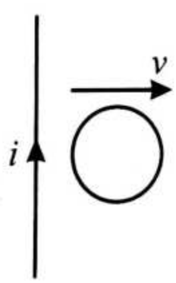
(1) 4 : 8 : 27 (2) 8 : 4 : 27
(3) 2 : 3 : 4 (4) 4 : 2 : 3

2. A particle starts moving on a circular path of radius 2m with time dependent speed $v = 2t$ m/s. Find the minimum time in which displacement of the particle becomes zero.



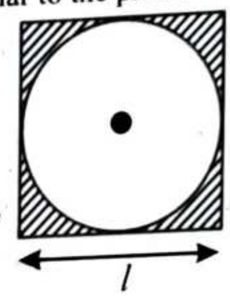
(1) $6\sqrt{\pi}$ s (2) $\sqrt{\pi}$ s
(3) $4\sqrt{\pi}$ s (4) $2\sqrt{\pi}$ s

3. A circular loop of radius r is moved away from a current carrying wire. The induced current in circular loop will be:



(1) clockwise
(2) anticlockwise
(3) first clockwise and then anticlockwise
(4) first anticlockwise and then clockwise

4. A square plate has uniformly distributed mass M and side l. A circular section of radius $\frac{l}{2}$ is removed from the plate. The center of the removed circle coincides with the center of the square as shown in the figure. Find the moment of inertia of the remaining plate about an axis passing through the common center and perpendicular to the plane.



(1) $\frac{Ml^2}{6}\left[1-\frac{\pi}{16}\right]$ (2) $\frac{Ml^2}{12}\left[1-\frac{3\pi}{16}\right]$
(3) $\frac{Ml^2}{6}\left[1-\frac{3\pi}{16}\right]$ (4) $\frac{Ml^2}{6}\left[1-\frac{\pi}{32}\right]$

5. A particle of mass M at rest decays into two particles of mass m_1 and m_2 having non-zero velocities. The ratio of de-Broglie wavelengths of the particles $\frac{\lambda_1}{\lambda_2}$ is:

(1) $\frac{m_1}{m_2}$ (2) $\frac{m_2}{m_1}$
(3) 1 : 1 (4) $\sqrt{\frac{m_2}{m_1}}$

6. When a photosensitive material is illuminated by light of frequency 3 times the threshold frequency, photoelectrons are emitted which give rise to a photocurrent. Now, if the frequency of light is reduced to one fifth of the original value but the intensity is tripled, then the photocurrent becomes:

(1) Quadrupled (2) Doubled
(3) Halved (4) Zero

7. If the speed of electron in the first Bohr orbit of hydrogen atom is 2.19×10^6 m/s, then the speed of the electron in the fifth orbit for He^+ is:
(1) 0.876×10^6 m/s (2) 1.095×10^6 m/s
(3) 1.523×10^6 m/s (4) 2.19×10^6 m/s

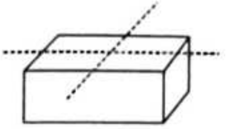
8. The binding energy of a nitrogen nucleus in MeV from the following data is:
$m_H = 1.00783u,\ m_n = 1.00876u$
(1) 100.6 MeV (2) 104.7 MeV
(3) 107.8 MeV (4) 109.6 MeV

9. The output voltage of an ideal transformer, connected to a 360 V A.C. mains is 36 V. When this transformer is used to light a bulb with rating 36 V, 36W, then the current in the primary coil of the circuit is:
(1) 0.01 A (2) 0.1 A
(3) 0.2 A (4) 1 A

10. A galvanometer having 60 divisions has current sensitivity of 10 μA/division. It has a resistance of 50 Ω. The value of shunt resistance to convert it into an ammeter which can measure currents up to 2 A is:
(1) 0.015 Ω (2) 0.15 Ω
(3) 1.5 Ω (4) 15 Ω

11. **Statement I:** The phase difference between electric and magnetic field in an *EM* wave is $\frac{\pi}{2}$.
Statement II: The oscillation of electric and magnetic field vectors are perpendicular to the direction of propagation of *EM* wave.
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

12. A bar magnet of length L and area of cross-section A is cut into four identical parts as shown in figure If initially magnetic moment was M, find magnetic moment of each piece after cutting:

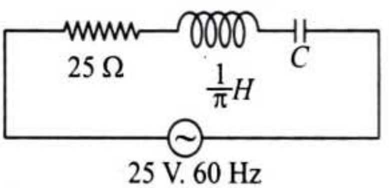


(1) $\frac{M}{4}$ (2) $\frac{M}{2}$
(3) $\frac{M}{8}$ (4) $2M$

13. Two thin lenses of focal length 2 m and 50 cm are placed in contact. Find equivalent power of combination of lenses.
(1) 0.4 D
(2) 2.5 D
(3) 0.52 D
(4) 0.025 D

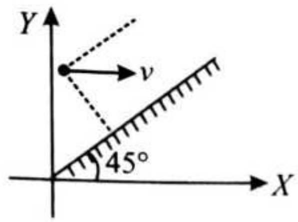
14. A series LCR circuit is connected across a source of emF $E = 20 \sin\left(50\pi t - \frac{\pi}{3}\right)$ V. If a current I = 10 sin ($50\pi t$) flows in the circuit, then the power dissipated through circuit is (t is in s):
(1) $50\sqrt{2}$ W
(2) 50 W
(3) $50\sqrt{3}$ W
(4) 100 W

15. An A.C. source of 25 V and frequency 60 Hz is applied across RLC series circuit as shown in the figure. If effective voltage across resistance is 25 V, then effective voltage across inductor is:



(1) 100 V (2) 120 V
(3) 60 V (4) 25 V

16. A plane mirror is inclined at an angle 45° in xy plane as shown in figure. A point object moves with speed v parallel to x-axis:



Find relative velocity of image w.r.t velocity of object.

(1) $v\hat{i}+v\hat{j}$　　(2) $v\hat{i}-v\hat{j}$

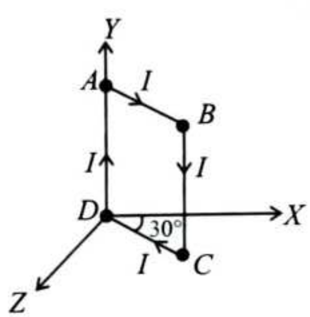
(3) $-v\hat{i}+v\hat{j}$　　(4) $-v\hat{i}-v\hat{j}$

17. Identify the correct relation between amplitude A and wave number k under which the maximum particle velocity is equal to three times that of the wave velocity corresponding to the equation: $y = A\sin(\omega t - kx)$.

(1) $A=\frac{\sqrt{3}}{k}$　　(2) $A=\frac{\sqrt{6}}{k}$

(3) $A=\frac{3}{k}$　　(4) $A=\frac{6}{k}$

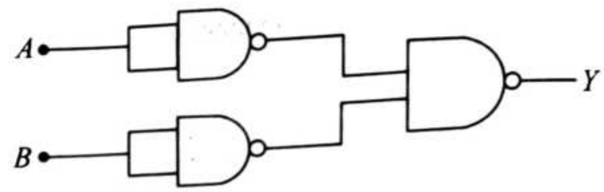
18. A square loop of side 'a' carries a current I. It is placed as shown in figure. Magnetic moment of the loop will be:



(1) $-\frac{Ia^2}{2}\left[\hat{i}+\sqrt{3}\hat{k}\right]$　　(2) $\frac{Ia^2}{2}\left[\hat{i}+\sqrt{3}\hat{k}\right]$

(3) $\frac{Ia^2}{2}\left[\hat{i}-\sqrt{3}\hat{k}\right]$　　(4) $-\frac{Ia^2}{2}\left[\hat{i}-\sqrt{3}\hat{k}\right]$

19. The truth table of given combination of gates shown is:



(1)

A	B	D
0	0	0
0	1	1
1	0	1
1	1	1

(2)

A	B	D
0	0	1
0	1	0
1	0	0
1	1	0

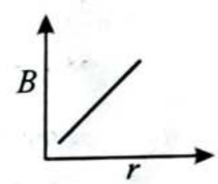
(3)

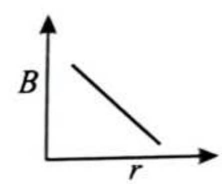
A	B	D
0	0	0
0	1	0
1	0	0
1	1	1

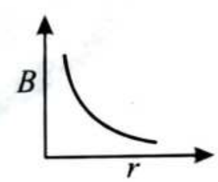
(4)

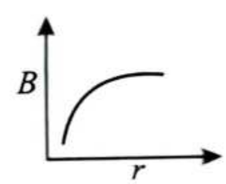
A	B	D
0	0	1
0	1	1
1	0	1
1	1	0

20. Which one of the following graphs shows the variation of magnetic induction B with distance r from a long wire carrying a current?

(1)


(2)


(3)


(4)


21. Two balls are projected upward simultaneously with speeds of 40 m/s and 60 m/s. Relative distance of the second ball with respect to the first ball at time t = 5 sec will be: [Neglect air resistance]

(1) 20 m

(2) 80 m

(3) 100 m

(4) 120 m

22. The radius of a circular current carrying coil is R. At what distance from the center of the coil on its axis, the intensity of magnetic field will be $\frac{1}{2\sqrt{2}}$ times that at the center?

(1) $2R$ (2) $\frac{3R}{2}$

(3) R (4) $\frac{R}{2}$

23. A projectile is given an initial velocity $\vec{u} = (\hat{i} + 3\hat{j})$ m/s where $\hat{i}$ is along the ground and $\hat{j}$ is along the vertical. The equation of trajectory of the particle is ($g = 10$ m/s^2)

(1) $y = x - 5x^2$ m

(2) $y = 3x - 5x^2$ m

(3) $6y = 2x - 5x^2$ m

(4) $3y = 6x - 25x^2$ m

24. The potential energy of a particle varies with distance x from a fixed origin as $U = \left(\frac{B\sqrt{x}}{x+A}\right)$, where A and B are constants. The dimensions of $\frac{B}{A}$ are:

(1) $[ML^2T^{-2}]$

(2) $[ML^{3/2}T^{-2}]$

(3) $[ML^{1/2}T^{-2}]$

(4) $[ML^{3/2}T^{-3/2}]$

25. The density of a sphere is measured by measuring its mass and diameter. If it is known that the maximum percentage errors in the measurement of mass and diameter arc 3% and 4% respectively, then find the maximum percentage error in the measurement of density.

(1) 18 % (2) 19 %

(3) 11 % (4) 15 %

26. The radius of newly discovered planet is 3 times that of the earth but having same density. If acceleration due to gravity on surface of the earth is g and that on surface of new planet is g', then:

(1) $g' = 2g$

(2) $g' = \frac{3}{2}g$

(3) $g' = 3g$

(4) $g' = 27g$

27. **Assertion (A):** On moving a distance two times of the initial distance, away from an infinitely long straight uniformly charged wire, the electric field is reduced to one third of its initial value.

Reason (R): The electric field is inversely proportional to the distance from an infinitely long straight uniformly charged wire.

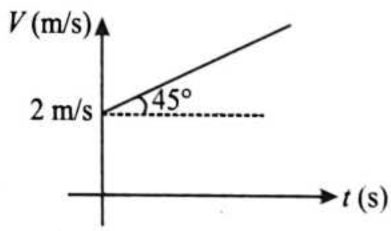
(1) Both **Assertion (A)** and **Reason (R)** are true and **Reason (R)** is a correct explanation of **Assertion (A)**.

(2) Both **Assertion (A)** and **Reason (R)** are true but **Reason (R)** is not a correct explanation of **Assertion (A)**.

(3) **Assertion (A)** is true and **Reason (R)** is false.

(4) Both **Assertion (A)** and **Reason (R)** are false

28. The velocity time graph of a particle moving on a straight line is shown below. What is the distance covered by the particle in 5th second of the motion?



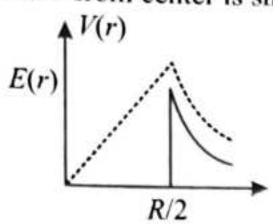
(1) 13.0 m

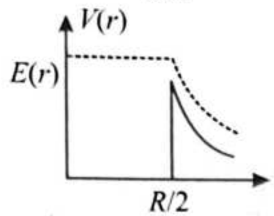
(2) 6.5 m

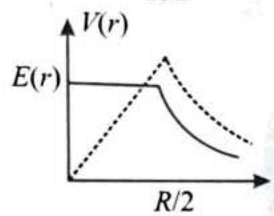
(3) 4.0 m

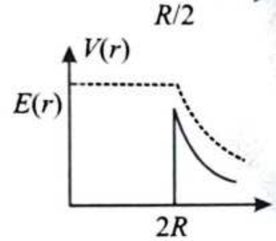
(4) 2.0 m

29. For a conducting spherical shell of radius $\frac{R}{2}$ with its center at origin, having uniform charge density. The variation of electric field $E(r)$ (Solid line) and potential and $V(r)$ (dotted line) with the distance r from center is shown as:

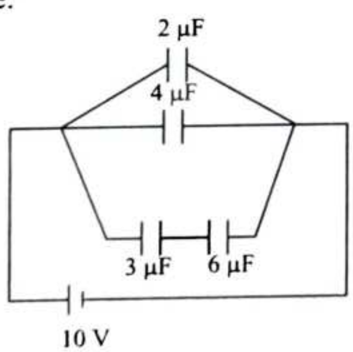
(1) 

(2) 

(3) 

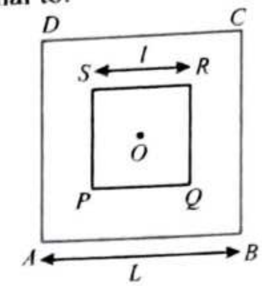
(4) 

30. Energy stored in the capacitor of capacitance 4 μF connected in the circuit as shown in figure will be:



(1) 200 μJ (2) 300 μJ
(3) 400 μJ (4) 800 μJ

31. A small square loop of wire of side ℓ is placed inside a large square loop of side L, ($L >> \ell$). If the loops are co-planar and their centers coincide, then mutual inductance of the system is directly proportional to:



(1) $\frac{L}{\ell}$

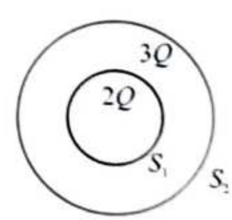
(2) $\frac{\ell}{L}$

(3) $\frac{L^2}{\ell}$

(4) $\frac{\ell^2}{L}$

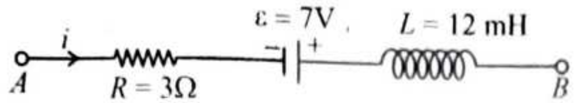
32. A projectile is thrown with speed 20 m/s at an angle 53° with the horizontal. Find the speed of the projectile when its velocity vector makes an angle 37° with the horizontal.
(1) 15 m/s
(2) 10 m/s
(3) 12 m/s
(4) 16 m/s

33. S_1 and S_2 are two concentric conducting shells having charges $2Q$ and $3Q$ respectively as shown in figure. The ratio of the electric flux through S_1 and S_2 is:



(1) 2 : 3 (2) 2 : 5
(3) 3 : 2 (4) 5 : 2

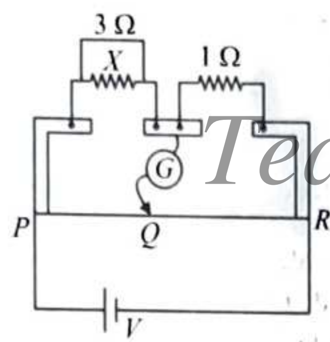
34. The electric network shown in figure is part of a complete circuit. If at any instant, current $i = 2$A and potential difference across A and B is zero, the value of $\left|\frac{di}{dt}\right|$ is:



(1) $\frac{26}{3}$ A/s (2) $\frac{25}{3}$ A/s

(3) $\frac{250}{3}$ A/s (4) $\frac{290}{3}$ A/s

35. In the meter bridge, galvanometer shows **zero** deflection if the balancing length $PQ = 30$ cm. The value of unknown resistance X is:

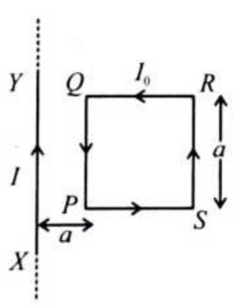


(1) 2 Ω (2) $\frac{1}{2}$ Ω

(3) 4 Ω (4) $\frac{1}{4}$ Ω

Physics : Section-B (Q. No. 36 to 50)

36. A square loop $PQRS$ carrying current I_0, is placed near and coplanar with a long straight conductor XY carrying current I. The net force on the loop will be:



(1) $\frac{\mu_0 I_0 I}{2\pi}$ attractive (2) $\frac{\mu_0 I_0 I}{4\pi}$ repulsive

(3) $\frac{\mu_0 I I_0}{2\pi}$ repulsive (4) $\frac{\mu_0 I I_0}{4\pi}$ attractive

37. Two moles of oxygen is mixed with ten moles of helium. The effective specific heat of the mixture at constant volume is: (Where R is universal gas constant)

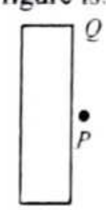
(1) 1 R

(2) 1.66 R

(3) 2 R

(4) 0.08 R

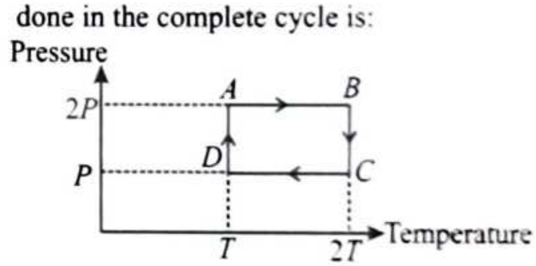
38. A charge Q is given to a neutral conducting plate of area A as shown below. The net electric field at point P which is located very near to the plate as shown in the figure is:



(1) $\frac{Q}{A\varepsilon_0}$ (2) $\frac{Q}{2A\varepsilon_0}$

(3) $\frac{Q}{4A\varepsilon_0}$ (4) $\frac{2Q}{A\varepsilon_0}$

39. One mole of an ideal gas having initial volume V pressure $2P$ and temperature T undergoes a cyclic process $ABCDA$ as shown below. The net work done in the complete cycle is:



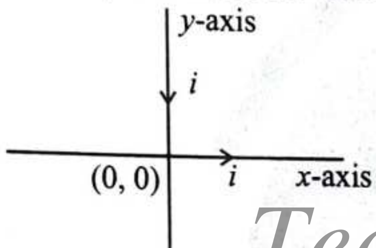
(1) Zero (2) $\frac{1}{2}RT\ell n2$

(3) $RT\ell n2$ (4) $\frac{3}{2}RT\ell n2$

40. A man measure the period of simple pendulum inside a stationary lift and the finds it T second. Now the lift accelerates upwards and the time period is $\frac{\sqrt{2}T}{3}$. The acceleration of the lift is: (Where g is acceleration due to gravity)

(1) $2g$ (2) $3g$
(3) $3.5g$ (4) $4.2g$

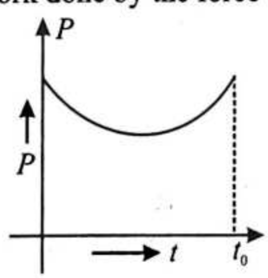
41. Equal currents are flowing in two infinitely long wires lying along x and y-axis in the direction shown in figure. Match the following two **List-I** (Magnetic field at a point) **List-II** (Direction of magnetic field) and choose the **correct** option.



List-I		List-II	
A.	Magnetic field at $(a, a, 0)$	I.	Perpendicular to z-axis
B.	Magnetic field at $(-a, -a, 0)$	II.	Along positive z-axis
C.	Magnetic field at $(a, -a, 0)$	III.	Along negative z-axis
D.	Magnetic field at $(0, 0, a)$	IV.	Zero

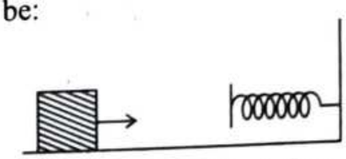
(1) A-I; B-II; C-III; D-IV
(2) A-II; B-III; C-IV; D-I
(3) A-III; B-I; C-II; D-IV
(4) A-II; B-IV; C-III; D-I

42. Power versus time graph for a force is given below. Work done by the force upto time t_0:



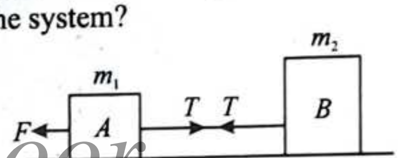
(1) First decreases then increases
(2) First increases then decreases
(3) Always increases
(4) Always decreases

43. A mass of 0.5 kg moving with a speed of 1.5 m/s on a horizontal smooth surface, collides with a nearly weightless spring of force constant $k = 50$ N/m. The maximum compression of the spring would be:



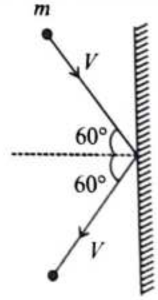
(1) 0.12 m (2) 1.5 m
(3) 0. 5 m (4) 0. 15 m

44. Which of the following is true about acceleration for the system?



(1) Acceleration is more in A, when force is applied on A.
(2) Acceleration is more in B, when force is applied on B.
(3) Acceleration is same and does not depend on whether the force is applied on m_1 or m_2.
(4) Acceleration depends on the tension in the string.

45. A rigid ball of mass m strikes a rigid wall at 60° and gets reflected without loss of speed as shown in the figure below. The value of impulse imparted by the wall on the ball will be:



(1) $\frac{mv}{2}$ (2) $\frac{mv}{3}$
(3) mv (4) $2mv$

[Contd...

46. A block of mass m is placed on a smooth wedge of inclination θ. The whole system is accelerated horizontally so that the block does not slip on the wedge. The force exerted by the wedge on the block (*g* is acceleration due to gravity) will be:

(1) $mg\sin\theta$ (2) mg

(3) $\frac{mg}{\cos\theta}$ (4) $mg\cos\theta$

47. A physical quantity of the dimensions of length that can be formed out of *c*, *G* and $\frac{e^2}{4\pi\varepsilon_0}$ is [*c* is velocity of light, *G* is universal constant of gravitational and *e* is charge]:

(1) $c^2\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{\frac{1}{2}}$ (2) $\frac{1}{c^2}\left[\frac{e^2}{G4\pi\varepsilon_0}\right]^{\frac{1}{2}}$

(3) $\frac{1}{c^2}G\frac{e^2}{4\pi\varepsilon_0}$ (4) $\frac{1}{c^2}\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{\frac{1}{2}}$

48. A body of mass *m* is attached to the lower end of a spring whose upper end is fixed. The spring has negligible mass. When the mass m is slightly pulled down and released, it oscillates with a time period of 3 s. When the mass m is increased by 1 kg, the time period of oscillations becomes 5 s. The value of *m* in kg is:

(1) $\frac{16}{9}$ (2) $\frac{9}{16}$

(3) $\frac{3}{4}$ (4) $\frac{4}{3}$

49. A wave travelling in the +*ve* *x*-direction having maximum displacement along *y*-direction as 1 m, wavelength 2π m and frequency of $\frac{1}{\pi}$ Hz is represented by:

(1) $y=\sin(2\pi x+2\pi t)$

(2) $y=\sin(x-2t)$

(3) $y=\sin(2\pi x-2\pi t)$

(4) $y=\sin(10\pi x-20\pi t)$

50. The equation of a simple harmonic wave is given by $y=3\sin\frac{\pi}{2}(50t-x)$ where *x* and *y* are in meter and *t* is in seconds. The ratio of maximum particle velocity to the wave velocity is:

(1) 2 π (2) $\frac{3\pi}{2}$

(3) 3 π (4) $\frac{2}{3\pi}$

Chemistry : Section-A (Q. No. 51 to 85)

51. Select the **correct** relation of E^o_{cell} with equilibrium constant (K_c) from the given options:

(1) $\frac{2.303RT}{nFE^o_{cell}}=\log K_c$

(2) $\frac{FE^o_{cell}}{2.303RT}=n\log K_c$

(3) $\frac{nFE^o_{cell}}{2.303RT}=\log K_c$

(4) $\frac{FE^o_{cell}}{RT}=2.303\,n\log K_c$

52. Spin only magnetic moment of V^{2+} ion is:

(1) $\sqrt{24}$ BM (2) $\sqrt{15}$ BM

(3) $\sqrt{8}$ BM (4) $\sqrt{2}$ BM

53. Which of the following compounds will show geometrical isomerism?

(1) But-2-ene

(2) 1,2-Dibromoethene

(3) But-2-ene-1,4-dioic acid

(4) All of these

54. Splitting up of the spectral lines in an electric field is called:

(1) Zeeman effect

(2) Stark effect

(3) Scattering effect

(4) Photoelectric effect

55. Which of the following laws of chemical combination is followed by CO_2 and CO?
(1) Law of conservation of mass
(2) Law of multiple proportions
(3) Law of constant proportions
(4) Gay Lussac's law of constant volume

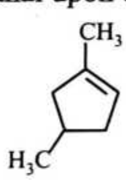
56. Methyl chloride when heated with silver cyanide, the major product formed is:
(1) $CH_3CH_2NH_2$
(2) CH_3CN
(3) CH_3NC
(4) $CH_3CH_2CH_2NH_2$

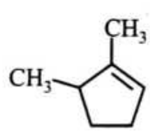
57. Which of the following is intensive property?
(1) Mass (2) Volume
(3) Pressure (4) Heat Capacity

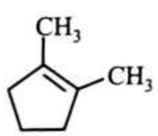
58. Which of the following hydrides of group 16 has highest boiling point?
(1) H_2Te (2) H_2O
(3) H_2Se (4) H_2S

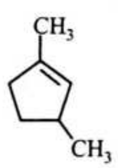
59. The complexes $[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$ are the examples of which type of isomerism:
(1) Linkage isomerism
(2) Ionisation isomerism
(3) Coordination isomerism
(4) Geometrical isomerism

60. Which compound would give 5-Keto-2-methyl hexanal upon ozonolysis?

(1) 

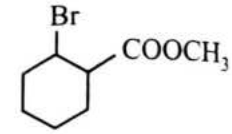
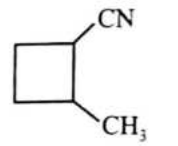
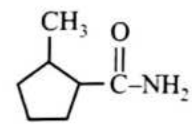
(2) 

(3) 

(4) 

61. Which of the following is water soluble vitamin?
(1) Vitamin C (2) Vitamin E
(3) Vitamin K (4) Vitamin A

62. Match **List-I** with **List-II:**

List-I (Name of the compound)		List-II (Structure)
A. 2-Methyl cyclopentane carboxamide	I.	
B. 2-Methyl cyclobutane carbonitrile	II.	
C. Cyclohexane carbonyl chloride	III.	
D. Methyl-2-bromocyclohexane carboxylate	IV.	

Choose the **correct** answer from the options given below:
(1) A-IV, B-I, C-II, D-III
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-I, C-III, D-IV
(4) A-II, B-III, C-IV, D-I

63. The energies E_1 and E_2 of two radiations are 75 eV and 300 eV respectively. The relations between their frequencies, i.e., ν_1 and ν_2 will be:
(1) $\nu_2 = 4\nu_1$ (2) $\nu_2 = 2\nu_1$
(3) $\nu_2 = \frac{1}{2}\nu_1$ (4) $\nu_2 = \nu_1$

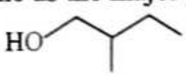
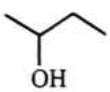
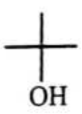
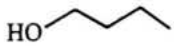
64. If we assume that $\Delta_0 <$ P.E for a particular octahedral complex, the **correct** representation of d^6 electronic configuration will be:
(1) $t_{2g}^3 e_g^3$ (2) $t_{2g}^4 e_g^2$
(3) $t_{2g}^6 e_g^0$ (4) $t_{2g}^5 e_g^1$

65. For the reaction $A + B \rightarrow C$, the rate is doubled if the concentration of A is doubled and is reduced by 2 if the concentration of B decreases by the factor of 4. The overall order of the reaction is:
(1) +1 (2) –0.5
(3) +3/2 (4) –5/2.5

66. The standard electrode potentials of four elements A, B, C and D are –3.10, 1.60, –0.50 and 0.80 volts respectively. The highest chemical reactivity is shown by:
(1) A (2) B
(3) C (4) D

67. $$CH_3CH_2CH_2CH_2OH \xrightarrow{K_2Cr_2O_7} A \xrightarrow[623\,K]{NaOH,\,CaO} B$$
The product 'B' in the above reaction is:
(1) $CH_3CH_2CH_2CH_3$
(2) $CH_3CH_2CH_2COOH$
(3) CH_3CH_3
(4) $CH_3CH_2CH_3$

68. The **incorrect** statement among the following is:
(1) The first ionisation potential of Al is less than the first ionisation potential of Mg.
(2) The third ionisation potential of Mg is greater than third ionisation potential of Na.
(3) The first ionisation potential of Na is less than the first ionisation potential of Mg.
(4) The second ionisation potential of Mg is greater than the second ionisation potential of Na.

69. Which alcohol on reaction with Cu at 573 K gives ketone as the major product?
(1) 
(2) 
(3) 
(4) 

70. K_1 and K_2 are equilibrium constant for reactions (I) and (II) then:
(I) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$
(II) $5NO(g) \rightleftharpoons \frac{5}{2}N_2(g) + \frac{5}{2}O_2(g)$
(1) $K_2 = \left(\frac{1}{K_1}\right)^5$ (2) $K_2 = (K_1)^5$
(3) $K_2 = \left(\frac{1}{K_1}\right)^{5/2}$ (4) $K_2 = (K_1)^{5/2}$

71. Given below are two statements:
Statement I: Gabriel phthalimide reaction gives aliphatic 1° amine only.
Statement II: Hinsberg's test does not respond to 3° amine.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

72. The oxidation state of I in $H_4IO_6^-$ is:
(1) +1 (2) –1
(3) +7 (4) +5

73. Equimolal aqueous and dilute solution of which compound will have the minimum freezing point?
(1) $K_4[Fe(CN)_6]$
(2) $[Ag(NH_3)_2]Cl$
(3) $[Co(NH_3)_6](NO_3)_3$
(4) $[Cu(NH_3)_4]SO_4$

74. For a reaction, $3A + 2B \rightarrow 5C$, the rate of appearance of C at time 't' is 4.5×10^{-7} mol L^{-1} s^{-1}. The rate of reaction is:

(1) 9×10^{6} mol L^{-1} s^{-1}
(2) 0.9×10^{-8} mol L^{-1} s^{-1}
(3) 9.0×10^{-8} mol L^{-1} s^{-1}
(4) 9.0×10^{-6} mol L^{-1} s^{-1}

75. Product (Y) of the given reaction is:

$$CH_3CH_2COOH \xrightarrow[\text{(ii) } H_2O]{\text{(i) } Cl_2/\text{Red P}} X \xrightarrow{\text{Alc. KOH}} Y$$

(1) $CH_3\underset{\displaystyle OH}{\underset{|}{C}}HCOOH$
(2) $\underset{\displaystyle OH}{\underset{|}{C}}H_2CH_2COOH$
(3) $CH_2=CHCOOH$
(4) $\underset{\displaystyle OH}{\underset{|}{C}}H_2\underset{\displaystyle OH}{\underset{|}{C}}HCOOH$

76. Assuming full decomposition, the volume of CO_2 released at STP on heating 39.4 g of $BaCO_3$ (at. mass of Ba = 137) will be:

(1) 4.48 L (2) 0.84 L
(3) 2.24 L (4) 4.96 L

77. Which one of the following options represents the **correct** bond order?

(1) $O_2^- > O_2 > O_2^+$ (2) $O_2^+ > O_2 > O_2^-$
(3) $O_2^+ > O_2^- > O_2$ (4) $O_2 > O_2^+ > O_2^-$

78. The standard enthalpy of formation is non-zero for:

(1) $Br_2(l)$ (2) S(rhombic)
(3) C(graphite) (4) C(diamond)

79. Calculate the vapour pressure of pure B, if $P_{total} = (0.195\chi_A + 0.315)$ atm at 295 K:

(1) 0.38 atm (2) 0.04 atm
(3) 0.315 atm (4) 0.172 atm

80. Match **List-I** with **List-II**:

List-I (Complex ion)		**List-II** (Number of unpaired electrons, hybridisation)	
A.	$[Cr(H_2O)_6]^{3+}$	I.	2, sp^3d^2
B.	$[Co(CN)_4]^{2-}$	II.	3, d^2sp^3
C.	$[Ni(NH_3)_6]^{2+}$	III.	5, sp^3d^2
D.	$[MnF_6]^{4-}$	IV.	1, dsp^2

Choose the **correct** answer from the options given below:

(1) A-III, B-I, C-II, D-IV
(2) A-III, B-II, C-IV, D-I
(3) A-II, B-I, C-III, D-IV
(4) A-II, B-IV, C-I, D-III

81. Given below are two statements:

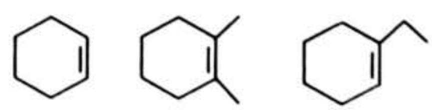
Statement I: In resonance, delocalisation of σ electrons takes place.

Statement II: In hyperconjugation, delocalisation of π electrons takes place.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

82. Arrange the following alkenes in increasing order of stability:



(I) (II) (III)

(1) I < II < III (2) II < I < III
(3) I < III < II (4) II < III < I

[Contd...

83. The number of significant figures in 0.00256 and 1.004 respectively are:

(1) 3, 2 (2) 5, 4
(3) 3, 4 (4) 6, 4

84. Which of the following has pOH is equal to near about one?

(1) $100\text{ mL }\frac{M}{10}\text{HCl}+100\text{ mL }\frac{M}{10}\text{NaOH}$

(2) $55\text{ mL }\frac{M}{10}\text{HCl}+44\text{ mL }\frac{M}{10}\text{NaOH}$

(3) $75\text{ mL }\frac{M}{10}\text{HCl}+25\text{ mL }\frac{M}{10}\text{NaOH}$

(4) $25\text{mL }\frac{M}{5}\text{ HCl}+75\text{mL }\frac{M}{5}\text{ NaOH}$

85. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: The outer electronic configuration of Cr and Cu are $3d^5 4s^1$ and $3d^{10} 4s^1$ respectively.

Reason R: Electrons are filled in orbitals in order of increasing energies given by $(n + l)$ rule.

In the In the light of the above statements, choose the **correct** answer from the options given below.

(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is not the correct explanation of **A**.

Chemistry : Section-B (Q. No. 86 to 100)

86. Lassaigne's test is not used for the detection of

(1) N (2) S
(3) Cl (4) O

87. Given below are two statements:

Statement I: Photoelectric effect shows that light has particle nature.

Statement II: Interference and diffraction show that light has wave nature.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

88. The pH of a 0.033 M solution of H_3PO_4 is:

(1) 0.099 (2) 1
(3) -1 (4) 0

89. The **correct** statement(s) regarding aryl halides is/are:

A. **Aryl halides** do not undergo Friedel-Crafts reaction.
B. Aryl halides are less reactive than haloalkanes towards nucleophilic substitution reactions.
C. Electron withdrawing substituents increases the rate of nucleophilic substitution in aryl halides.

(1) A and B only
(2) B only
(3) B and C only
(4) A, B and C

90. Match **List-I** with **List-II**:

	List-I (Element)		List-II (Group number)
A.	N	I.	2
B.	Be	II.	3
C.	Sc	III.	17
D.	Cl	IV.	15

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-IV, D-III
(2) A-II, B-I, C-III, D-IV
(3) A-II, B-I, C-IV, D-III
(4) A-IV, B-I, C-II, D-III

91. What is the shape of ClF_3 molecule?
(1) Trigonal bipyramidal
(2) T-shape
(3) Trigonal planar
(4) Pyramidal

92. Given below are two statements:
Statement I: Bi^{3+} is more stable than Bi^{5+}.
Statement II: Due to inert pair effect, lower oxidation state of Bi is more stable.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

93. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: o-nitrophenol and p-nitrophenol can be separated by steam distillation.
Reason R: o-nitrophenol is steam volatile whereas p-nitro phenol is not steam volatile.
In the In the light of the above statements, choose the **correct** answer from the options given below.
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is not the correct explanation of **A**.

94. $C_6H_6(l)+\frac{15}{2}O_2(g)\rightarrow 6CO_2(g)+3H_2O(l)$
The above reaction at 27° proceeds spontaneously because the magnitude of:
(1) $\Delta H = T\Delta S$ (2) $\Delta H > T\Delta S$
(3) $\Delta H < T\Delta S$ (4) $\Delta H > 0$ and $T\Delta S < 0$

95. Aldehydes can be prepared by Etard's reaction. The reagent used in Etard's reaction is:
(1) CrO_2Cl_2 (2) Cr_2O_3
(3) $Cr_2O_7^{2-}$ (4) K_2CrO_4

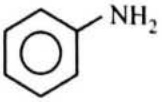
96. $\text{Phenol} \xrightarrow[\text{(ii) } CCl_4]{\text{(i) NaOH}} B \xrightarrow[\text{(iv) HCl}]{\text{(iii) NaOH}} C \xrightarrow{(CH_3CO)_2O} D$
The **correct** IUPAC name of product 'D' in the above reaction is:
(1) Salicylaldehyde
(2) 2-Hydroxybenzoic acid
(3) Salicylic acid
(4) 2-Acetoxybenzoic acid

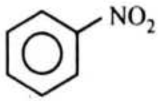
97. The **correct** IUPAC name of the given compound is:
$[Pt(Py)_4][PtCl_4]$
(1) Tetrapyridineplatinum(II)tetrachloridoplatinate(II)
(2) Tetrapyridineplatinum(II)tetrachloridoplatinum(II)
(3) Tetrapyridineplatinum(II)tetrachloridoplatinum(IV)
(4) Tetrapyridineplatinum(IV)tetrachloridoplatinum(IV)

98. The electronic configuration of few elements is given below. Mark the statement which is **not** correct about these elements.
I. $1s^22s^22p^63s^1$
II. $1s^22s^22p^5$
III. $1s^22s^22p^6$
IV. $1s^22s^22p^3$
(1) (I) is an alkali metal
(2) (III) is a noble gas
(3) (II) is a non-metal
(4) (IV) has high metallic character than (I)

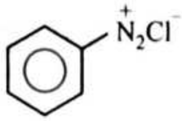
99. The **correct** increasing order of reducing power of dipositive ions of lanthanoid is:
(1) $Sm^{2+} < Yb^{2+} < Eu^{2+}$
(2) $Eu^{2+} < Yb^{2+} < Sm^{2+}$
(3) $Sm^{2+} < Eu^{2+} < Yb^{2+}$
(4) $Yb^{2+} < Eu^{2+} < Sm^{2+}$

100. Which of the following compound will be suitable for Kjeldahl's method for nitrogen estimation?

(1) 

(2) 

(3) 

(4) 

Botany : Section-A (Q. No. 101 to 135)

101. Function(s) of the cell wall is/are:

(1) provides shape to the cell.
(2) helps in cell-to-cell interaction.
(3) provides barrier to undesirable macromolecules.
(4) All of these.

102. When a dot is present at the top of floral diagram, what does it indicate?

(1) Position of mother axis.
(2) Position of gametes.
(3) Fusion of whorls.
(4) Nature of flower.

103. How many types of gametes will be produced by an individual with AABbcc genotype?

(1) Two (2) Four
(3) Six (4) Eight

104. How many chromosomes will the cell have at G_1 phase, after S phase, and after M phase respectively, if it has 14 chromosomes at metaphase?

(1) 7, 14 and 28 respectively
(2) 14, 14 and 14 respectively
(3) 14, 28 and 14 respectively
(4) 14, 7 and 14 respectively.

105. Fluid mosaic model was an improved model of the structure of cell membrane proposed by;

(1) Singer and Robert Brown.
(2) Anton Von Leeuwenhoek and Palade.
(3) Nicolson and Anton Von Leeuwenhoek.
(4) Singer and Nicolson.

106. Read the following and select the **incorrect** statement.

(1) Category is a unit of classification.
(2) Classification is a single step process.
(3) Category represents rank.
(4) Rank is commonly termed as taxa.

107. Given below are two statements:

Statement I: Slime moulds are saprophytic protists.

Statement II: Under unfavourable conditions, slime moulds form an aggregation called *plasmodium* which may grow and spread over several feet.

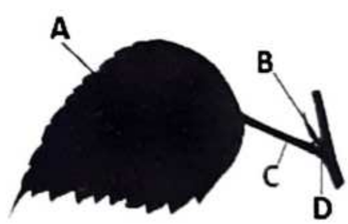
In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

108. In pteridophytes, sporophyte and gametophyte are respectively;

(1) independent differentiated body and independent thalloid body.
(2) dependent differentiated body and independent thalloid body.
(3) independent thalloid body and independent differentiated body.
(4) dependent thalloid body and independent differentiated body.

109. Select the option which represents labeled part (**A, B, C** and **D**) correctly.



(1) A-Lamina; B-Stipule; C- Petiole; D-Axillary bud
(2) A-Stipule; B-Lamina; C-Petiole; D-Axillary bud
(3) A-Stipule; B-Petiole; C-Lamina; D-Axillary bud
(4) A-Lamina; B-Petiole; C-Stipule; D-Axillary bud

110. All the following options are true for VNTRs, **except** that they;
(1) show a very low degree of polymorphism.
(2) are a type of mini-satellite DNA.
(3) are used as a probe in DNA fingerprinting.
(4) have a size range from 0.1 to 20 kb.

111. Choose the **incorrect** match from the following.

(1) Nucleus	-	RNA
(2) Lysosome	-	Protein synthesis
(3) Mitochondria	-	Aerobic respiration
(4) Cytoskeleton	-	Microtubules

112. Members of phycomycetes are found:
A. in aquatic habitats.
B. on decaying wood.
C. in moist and damp places.
D. as obligate parasites on plants.
Choose the **correct** option.
(1) A and B only
(2) C only
(3) B and D only
(4) All are correct

113. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The respiratory pathway is considered as an amphibolic pathway rather than as a catabolic one.
Reason R: During breakdown and synthesis of complex molecules like fatty acids and proteins, respiratory intermediates form the link for it.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

114. Which of the following is/are produced by plant as defences against grazers and browsers:
A. nicotine.
B. caffeine.
C. quinine.
D. strychnine.
Choose the *most appropriate* answer from the options given below:
(1) (B), (C) and (D) only
(2) (B) and (C) only
(3) (A) only
(4) (A), (B), (C) and (D)

115. When F_2 generation in a Mendelian cross shows that both genotypic and phenotypic ratios are same as 1 : 2 : 1 then it represents a case of:
(1) monohybrid cross with incomplete dominance.
(2) polygenic inheritance.
(3) dihybrid cross.
(4) monohybrid cross with complete dominance.

116. Cellular energy in the form of ATP is formed in:
(1) cell wall. (2) ribosomes.
(3) mitochondria. (4) Golgi bodies.

117. Which of the following is a **mismatched** pair?
(1) Mutualism - Both species are benefited.
(2) Commensalism- One species is benefited
(3) Ammensalism- Both species are harmed.
(4) Predation-One species is harmed.

118. Select the **incorrect** pair from the following w.r.t. percentage of dwarf plants obtained during monohybrid cross in F_1 generation.

(1) Tt × tt - 50 percent
(2) Tt × Tt - 25 percent
(3) TT × tt - 50 percent
(4) tt × tt - 100 percent

119. Match **List-I** and **List-II**.

List-I (Group of bacteria)	List-II (Their shape)
(A) *Coccus*	(I) Rod-shaped
(B) *Bacillus*	(II) Spherical
(C) *Spirillum*	(III) Spiral
(D) *Vibrium*	(IV) Comma-shaped

Choose the **correct** answer from the option given below:

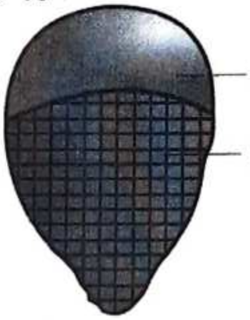
(1) A-I, B-II, C-III, D-IV
(2) A-II, B-I, C-III, D-IV
(3) A-I, B-III, C-IV, D-II
(4) A-IV, B-III, C-II, D-I

120. Which of the following are the uses of pteridophytes?

A. Medicinal purposes
B. Soil-binders
C. Ornamentals
D. Commercial production of hydrocolloids

(1) A and D (2) C and D
(3) A, B and D (4) A, B and C

121. Identify type of vascular bundle in given figure-



(1) Conjoint open
(2) Radial
(3) Conjoint radial
(4) Conjoint closed

122. Identify the **incorrect** statement from the following w.r.t megasporogenesis in flowering plants.

(1) The process of formation of megaspores from the megaspore mother cell is called megasporogenesis.
(2) Ovules generally differentiate a single megaspore mother cell (MMC) in the micropylar region of the nucellus.
(3) MMC is a large cell containing dense cytoplasm and a prominent nucleus.
(4) The MMC undergoes mitotic division that results in the production of four megaspores.

123. All of the following are stop codons, **except**;

(1) UAA. (2) UAG.
(3) UGA. (4) UGG.

124. Choose the **incorrect** pair from the following.

(1) Karyokinesis - Cell plate formation
(2) Cytokinesis - Division of cytoplasm
(3) Prophase - Aster formation
(4) Centrioles - Duplicates in the cytoplasm

125. Match **List-I** with **List-II**.

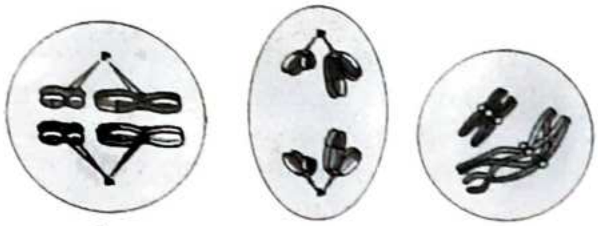
List-I	List-II
(A) Primary sludge	(I) Masses of bacteria ssociated with fungal filament
(B) Effluent	(II) Solids settled after sequential filtration
(C) Flocs	(III) Supernatant
(D) Toddy	(IV) A traditional drink formed after fermentation

Choose the **correct** answer from the options given below:

(1) A-II, B-I, C-III, D-IV
(2) A-II, B-III, C-I, D-IV
(3) A-III, B-II, C-I, D-IV
(4) A-I, B-III, C-II, D-IV

[Contd...

126. A, B and C are three stages in the meiosis-I as given in diagrams:



A B C

The **correct** sequence of events that occurs during meiosis I is:

(1) A → C → B. (2) A → B → C.

(3) C → A → B. (4) C → B → A.

127. Identify the **correct** features of mango and coconut fruits.

A. In both fruit is a drupe.

B. Endocarp is edible in both.

C. Mesocarp in Coconut is fibrous, and in Mango it is fleshy.

D. In both, fruit develops from monocarpellary ovary.

Choose the *most appropriate* answer from the options given below:

(1) A, B, and C only (2) A, and D only

(3) A and B only (4) A, C and D only

128. The Central dogma in molecular biology states that the genetic information flows from:

(1) DNA → RNA → Protein.

(2) RNA → DNA → Protein.

(3) DNA → Protein → RNA.

(4) Protein → DNA → RNA.

129. Identify **wrong** statements from the following.

A. Yeast can progress through the cell cycle in only about 90 minutes.

B. Spindle fibres attach to square shaped kinetochores during metaphase.

C. There is no replication of DNA during interkinesis.

D. Terminalisation of chiasmata occurs in Anaphase-I.

Choose the *most appropriate* answer from the options given below:

(1) B and D only (2) C and D only

(3) A and B only (4) A and D only

130. Match **List-I** with **List-II**:

	List-I		List-II
(A)	Compositae	(I)	Cotton
(B)	Gramineae	(II)	Cabbage
(C)	Malvaceae	(III)	Sunflower
(D)	Cruciferae	(IV)	Wheat

Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-I, D-II

(2) A-III, B-IV, C-I, D-II

(3) A-III, B-I, C-IV, D-II

(4) A-IV, B-I, C-III, D-II

131. Penicillin was the first antibiotic which was a chance discovery. Alexander Fleming at that time was working on;

(1) *Streptococci* bacteria.

(2) *Penicillium pneumonia.*

(3) *Penicillium notatum.*

(4) *Staphylococci* bacteria.

132. Which of the following are primary producers in an aquatic ecosystem?

(a) Phytoplankton

(b) Zooplankton

(c) Fishes

(d) Algae

(e) Aquatic plants

Options:

(1) (a), (b) and (c) only

(2) (b), (c) and (d) only

(3) (a), (d) and (e) only

(4) (a) and (c) only

133. In an ecosystem energy enters through:

(1) herbivores. (2) carnivores.

(3) producers (4) decomposers.

134. Select the **mis-matched** pair.

(1)	Pleiotropy	-	Multiple phenotypic expressions of a gene
(2)	Test cross	-	Cross of a progeny with recessive parent
(3)	Polygenic traits	-	Controlled by one gene
(4)	Incomplete dominance	-	Flower colour in snapdragon

135. Arrange the following stages in the developmental process of a plant cell from meristematic cell to the onset of senescence:

(a) Differentiation
(b) Maturation
(c) Meristematic cell
(d) Senescence
(e) Plasmatic growth
(f) Expansion (Elongation)

Options:

(1) (a)-(b)-(c)-(d)-(e)-(f)
(2) (b)-(c)-(d)-(e)-(f)-(a)
(3) (d)-(c)-(b)-(a)-(f)-(e)
(4) (c)-(e)-(f)-(a)-(b)-(d)

Botany : Section-B (Q. No. 136 to 150)

136. Match **List-I** with **List-II** to find out the **correct** option w.r.t structure of pollen grain.

	List-I		List-II
(A)	The inner wall	(I)	Bigger cell
(B)	The outer wall	(II)	Smaller cell
(C)	The vegetative cell	(III)	Exine
(D)	The generative cell	(IV)	Intine

	A	B	C	D
(1)	II	IV	I	III
(2)	IV	III	I	II
(3)	I	IV	II	III
(4)	III	IV	II	I

137. Which of the following is **not** a reason for Mendel's success?

(1) Large sampling size.
(2) Selection of true breeding pea plants with contrasting traits.
(3) Explained incomplete dominance in genes.
(4) Use of mathematics and statistics.

138. From the given set of structures which are found in mitochondria.

A. 70S ribosomes.
B. 80S ribosomes.
C. Single circular DNA.
D. Few RNA molecules.
E. Stroma lamellae.

Choose the *most appropriate* answer from the options given below:

(1) B, D and E only
(2) A and E only
(3) A, D and E only
(4) A, C and D only

139. Given below are two statements:

Statement I: Majority of flowering plants produce hermaphrodite flowers.

Statement II: Continued self-pollination results in inbreeding depression.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

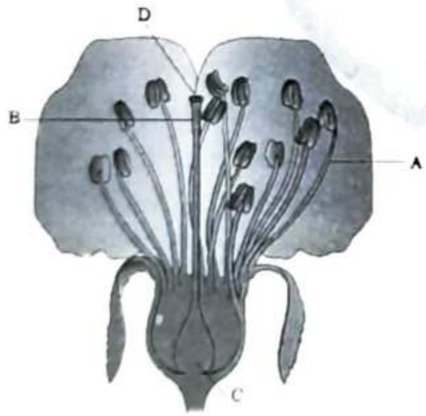
140. The given diagram refers to;

(1) a hypogynous flower.
(2) a perigynous flower.
(3) an epigynous flower.
(4) none of these.

141. Choose the **incorrect** option.

(1) Sucrose is converted into glucose and fructose by enzyme, intvertase.
(2) In EMP, hexose sugar splits into two molecule of triose.
(3) EMP is oxygen dependent pathway.
(4) Two ATP are required for the breakdown of glucose

142. Identify **A**, **B**, **C** and **D** from the given figure and select the **correct** option.



	A	B	C	D
(1)	Filament	Style	Ovary	Stigma
(2)	Style	Filament	Ovary	Stigma
(3)	Filament	Stigma	Ovary	Style
(4)	Stigma	Style	Ovary	Filament

143. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The DNA strands in a double helix are said to be complementary to each other.

Reason R: If the sequence of bases in one strand of DNA is known then the sequence in the other strand cannot be predicted.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

144. Select **incorrect** for Punnett Square.

(1) It was developed by a Greek geneticist.
(2) It is a graphical representation.
(3) It helps to calculate the probability of all possible genotypes of offspring in a genetic cross.
(4) The possible gametes are written on two sides, usually the top row and left columns.

145. For synthesis of a molecule of glucose by a C_3 plant, the number of molecules of NADPH and ATP required is respectively;

(1) 15 and 10.
(2) 12 and 18.
(3) 12 and 6.
(4) 18 and 12.

146. Increased diversity contributed to higher productivity was shown through experiments by:

(1) Paul Ehrlich
(2) Alexander von Humboldt
(3) David Tilman
(4) Robert May

147. Match the **List-I** with **List-II** w.r.t *lac* operon.

	List-I		**List-II**
(A)	*i* gene	(I)	Codes for beta-galactosidase
(B)	*z* gene	(II)	Codes for transacetylase
(C)	*y* gene	(III)	Codes for the repressor
(D)	*a* gene	(IV)	Codes for permease

Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-IV, D-I
(2) A-I, B-IV, C-III, D-II
(3) A-III, B-I, C-IV, D-II
(4) A-III, B-II, C-I, D-IV

148. Find the **wrongly** matched pair.

(1) Endemism - Species confined to a region and not found anywhere else
(2) Biodiversity Hotspots - Western ghats
(3) Sacred groves - Jaintia hills of Rajasthan
(4) *Ex-situ* conservation - Wildlife safari parks

149. Which of the following pair are **correctly** matched?

A. Trichome - Fundamental tissue system
B. Guard cells - Conducting tissue system
C. Xylem - Epidermal tissue system
D. Root hair - Epidermal tissue system

Choose the *most appropriate* answer from the options given below:

(1) B and D only
(2) A, C and D only
(3) D only
(4) A, B and C only

150. How many components are associated with non-cyclic photophosphorylation?

Water splitting, P700, P680, ETS, RuBisCO, PEPcase, NADP reductase

(1) Three (2) Two
(3) Four (4) Five

Zoology : Section-A (Q. No. 151 to 185)

151. In India, population crossed one billion mark in 2000. The probable reasons for this are, decline in:

A. maternal mortality rate
B. infant mortality rate
C. number of people in reproducible age
D. death rate

Choose the *most appropriate* answer from the options given below.

(1) A and B only
(2) A, B and C only
(3) A, B and D only
(4) A, B, C and D

152. Match **List-I** with **List-II**

	List-I		**List-II**
(A)	**Alkaloids**	(I)	Vinblastin
(B)	**Essential** oils	(II)	Morphine
(C)	**Toxins**	(III)	Lemon grass oil
(D)	**Drugs**	(IV)	Abrin

Choose **the correct** answer from the options given below:

(1) A-II, B-III, C-IV, D-I
(2) A-I, B-IV, C-II, D-III
(3) A-IV, B-III, C-II, D-I
(4) A-III, B-I, C-II, D-IV

153. Which of the following statements are **correct** regarding phylum Aschelminthes?

A. The body is circular in cross-section hence the name roundworms.
B. Alimentary canal is complete with a well-developed muscular pharynx.
C. Sexes are separate (dioecious), i.e., males and females are distinct.
D. Nephridia help in osmoregulation and excretion.

(1) A and B only (2) C and D only
(3) A, B and C only (4) All of these

154. Match **List-I** with **List-II**.

	List-I		**List-II**
(A)	Larynx	(I)	Lid of larynx
(B)	Trachea	(II)	Air sacs
(C)	Alveoli	(III)	Voice box
(D)	Epiglottis	(IV)	Wind pipe

Choose the **correct** answer from the options given below:

(1) A-III, B-I, C-II, D-IV

(2) A-III, B-IV, C-I, D-II

(3) A-I, B-II, C-III, D-IV

(4) A-III, B-IV, C-II, D-I

155. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: In both human male and female, reproductive systems are completely located in pelvis region.

Reason R: Testes and ovaries are primary sex organs as they produce gametes.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) **A** is true but **R** is false.

(2) **A** is false but **R** is true.

(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.

(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

156. Which of the following is **not** an example of homologous organs?

(1) Forelimbs of a whale and a bat.

(2) Wings of a butterfly and a bird.

(3) Vertebrate heart

(4) Human arm and the whale's flipper.

157. Given below are two statements:

Statement I: Enzymes are proteinaceous in nature.

Statement II: All the carbon compounds that we get from living tissues can be called biomolecules.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

158. Muscle fibres with fusiform ends are:

(1) cardiac muscle fibres.

(2) smooth muscle fibres.

(3) skeletal muscle fibres.

(4) present in biceps.

159. Match **List-I** with **List-II**.

	List-I		**List-II**
(A)	Autonomic neural system	(I)	Corpora quadrigemina
(B)	Forebrain	(II)	Transmit impulse from CNS to involuntary organs.
(C)	Synapse	(III)	Cerebrum, thalamus, hypothalamus
(D)	Midbrain	(IV)	Junction through which impulse is transmitted.

Choose the **correct** answer from the options given below:

(1) A-I, B-III, C-II, D-IV

(2) A-III, B-I, C-IV, D-II

(3) A-II, B-III, C-IV, D-I

(4) A-I, B-II, C-III, D-IV

160. Given below are two statements:

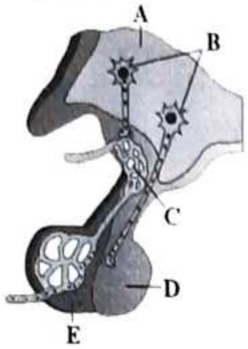
Statement I: Tertiary structure is necessary for many biological activities of proteins.

Statement II: Antibodies are proteins.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

161. Observe the given diagrammatic representation. Identify **A** to **E**:



(1) **A**-Hypothalamus; **B**-Hypothalamic neurons; **C**-Portal circulation; **D**-Posterior pituitary, **E**- Anterior pituitary.
(2) **A**-Hypothalamic neurons; **B**-Hypothalamic veins; **C**-Pars distalis, **D**-Pars intermedia; **E**- Hypothalamus.
(3) **A**-Hypothalamic neurons; **B**-Portal circulation; **C**-Anterior pituitary, **D**-Posterior pituitary; **E**-Hypothalamic vein.
(4) **A**-Hypothalamic neurons; **B**-Portal circulation; **C**-Posterior pituitary; **D**-Anterior pituitary; **E**-Hypothalamus.

162. Which of the following is a primary lymphoid organ?
(1) Spleen (2) Lymph nodes
(3) Thymus (4) Tonsils

163. Graves' disease is caused due to:
(1) hyposecretion of thyroid gland.
(2) hypersecretion of thyroid gland.
(3) hyposecretion of adrenal gland.
(4) hypersecretion of adrenal gland.

164. In the context of biotechnology, what does "GMO" stand for?
(1) Genetically Modified Organism
(2) Gene Manipulated Organism
(3) Genetic Mapped Organism
(4) General Modified Organism

165. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The macrophage of areolar tissue secretes collagen fibres.
Reason R: The fibres provide strength, elasticity and flexibility to the tissue.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

166. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Lamarck	(I)	Embryological support for evolution
(B)	Ernst Heckel	(II)	Inheritance of acquired characters
(C)	Alfred Wallace	(III)	Essay on population
(D)	Malthus	(IV)	Co-proposer of natural selection

Choose the **correct** answer from the options given below:
(1) A-I, B-II, C-IV, D-III
(2) A-II, B-I, C-III, D-IV
(3) A-II, B-I, C-IV, D-III
(4) A-III, B-II, C-I, D-IV

167. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Plasma	(I)	Osmotic balance
(B)	Formed elements	(II)	Blood clotting
(C)	Albumin	(III)	55% of blood volume
(D)	Fibrinogens	(IV)	45% of the blood

[Contd...

Choose the **correct** answer from the options given below:

(1) A-II, B-I, C-III, D-IV
(2) A-III, B-IV, C-I, D-II
(3) A-IV, B-III, C-I, D-II
(4) A-I, B-II, C-III, D-IV

168. Natural killer cell represents which of the following type of immune cells in our body?

(1) Thrombocytes
(2) Stem cells
(3) Lymphocytes
(4) Polymorpho-nuclear leukocytes

169. Choose the **correct** order of the reproductive tract of a male human beings.

(1) Rete testis → Epididymis → Vasa efferentia → Urethra → Vas deferens.
(2) Rete testis → Vasa efferentia → Epididymis → Vas deferens → Urethra.
(3) Vasa efferentia → Epididymis → Vas deferens → Urethra → Rete testis.
(4) Urethra → Rete testis → Vas deferens → Epididymis → Vasa efferentia.

170. The nerve centres which control the body temperature and the urge for eating are contained in:

(1) hypothalamus.
(2) pons.
(3) cerebellum.
(4) thalamus.

171. Ionizing radiations like X-rays and gamma rays and non-ionizing radiations like UV cause DNA damage leading to neoplastic transformation are called as:

(1) carcinogens.
(2) oncogenes.
(3) pollutants.
(4) mutagens.

172. Given below are two statements:

Statement I: Lactational amenorrhea is considered effective upto 24 weeks following parturition.

Statement II: ZIFT involves the transfer of zygote or early embryos with upto 8 blastomeres into the fallopian tube.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

173. In the field of biotechnology, 'molecular scissors' refer to:

(1) ligases.
(2) polymerases.
(3) restriction enzymes.
(4) Other genetic engineering tools.

174. Nematode-specific genes were introduced into the host plant by:

(1) Retrovirus.
(2) Adenovirus.
(3) *Meloidegyne incognitia.*
(4) *Agrobacterium.*

175. Which of the following is a component of nucleotides, the building blocks of nucleic acids?

(1) Amino acids
(2) Fatty acids
(3) Nitrogenous bases
(4) Polysaccharides

176. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Lymphatic system	(I)	Carries oxygenated blood
(B)	Pulmonary vein	(II)	Immune response
(C)	Thrombocytes	(III)	To drain back the tissue fluid to the circulatory system
(D)	Lymphocytes	(IV)	Coagulation of blood

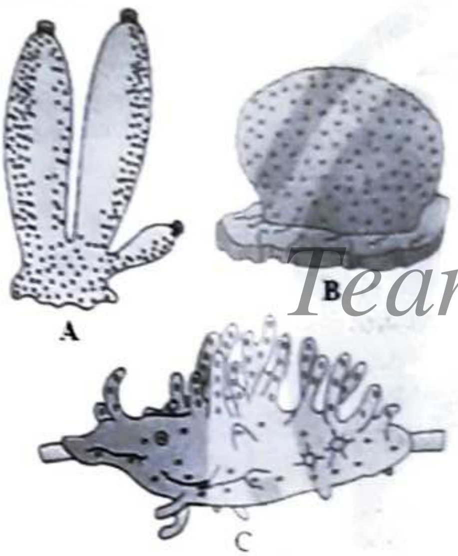
Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-III, D-IV
(2) A-III, B-I, C-IV, D-II
(3) A-II, B-I, C-III, D-IV
(4) A-I, B-III, C-II, D-IV

177. Which of the following is the major component of oral contraceptive pills?

(1) Copper ions
(2) Growth hormone
(3) Progesterone
(4) Luteinizing hormone

178. Examine the figures **A**, **B** and **C**. In which one of the four options all the items, **A**, **B** and **C** **are correct**?



	A	B	C
(1)	*Sycon*	*Euspongia*	*Spongilla*
(2)	*Euspongia*	*Spongilla*	*Sycon*
(3)	*Spongilla*	*Sycon*	*Euspongia*
(4)	*Euspongia*	*Sycon*	*Spongilla*

179. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: *Salmonella typhi* causes typhoid fever in human beings.

Reason R: Normal low fever, weakness, stomach pain, constipation, headache and loss of appetite are some of the common symptoms of this disease.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

180. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Mons pubis	(I)	Tiny finger like structure which lies at the upper junction of two labia minora.
(B)	Labia majora	(II)	Paired folds of tissue under the labia majora.
(C)	Labia minora	(III)	Paired fleshy folds of tissue which extend down from mons pubis.
(D)	Clitoris	(IV)	Fatty tissue covered by skin and pubic hairs.

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-III, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-IV, C-I, D-III
(4) A-III, B-IV, C-I, D-II

181. Joint between carpal and metacarpal of human thumb is:

(1) ball and socket joint.
(2) hinge joint.
(3) saddle joint.
(4) gliding joint.

182. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Adaptive radiation	(I)	Selection of resistant varieties due to excessive use of herbicides and pesticides
(B)	Convergent evolution	(II)	Bones of forelimbs in man and whale
(C)	Divergent evolution	(III)	Flippers of Penguins and Dolphins
(D)	Evolution by anthropogenic action	(IV)	Darwin finches

Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-I, D-IV
(2) A-I, B-II, C-III, D-IV
(3) A-I, B-IV, C-III, D-II
(4) A-IV, B-III, C-II, D-I

183. Golden Rice is a genetically modified variety of rice that contains which vitamin?

(1) Vitamin A (2) Vitamin C
(3) Vitamin D (4) Vitamin B_{12}

184. The **correct** sequence of the parts of alimentary canal in cockroach is:

(1) mouth → pharynx → crop → oesophagus → gizzard → midgut → caecum → colon → rectum → anus.
(2) mouth → pharynx → oesophagus → gizzard → crop → midgut → ileum → colon → rectum → anus.
(3) mouth → pharynx → oesophagus → crop → proventriculus → midgut → ileum → colon → rectum → anus.
(4) mouth → pharynx → oesophagus → stomach → crop → midgut → colon → caecum → rectum → anus.

185. Which of the following hormones does **not** interact with intracellular receptors?

(1) Cortisol (2) Iodothyronine
(3) Glucagon (4) Estradiol

Zoology : Section-B (Q. No. 186 to 200)

186. A bundle of fibres present in the myocardium which helps in the transmission of cardiac impulses from atrioventricular node, to ventricles is:

(1) SAN
(2) pacemaker.
(3) bundle of His.
(4) papillary muscles.

187. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The pharynx opens through the larynx region into the trachea.

Reason R: Larynx is a cartilaginous box which helps in sound production.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

188. Which of the statements is **incorrect**?

(1) Population of India was approximately 150 million at the time of our independence.
(2) Population reached the mark of 1.2 billion by May, 2011.
(3) 'Family planning' program was initiated in 1951.
(4) Successful implementation of various action plans to attain reproductive health requires strong infrastructural facilities, professional expertise and material support.

189. Match **List-I** with **List-II**.

	List-I		List-II
(A)	*cry I Ab*	(I)	Transgenic animal
(B)	Urine analysis	(II)	Corn borer
(C)	Rosie	(III)	Cotton bollworms
(D)	*cry II Ab*	(IV)	Conventional method of diagnosis

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-IV, D-I
(2) A-II, B-IV, C-I, D-III
(3) A-II, B-I, C-IV, D-III
(4) A-I, B-II, C-III, D-IV

190. Which of the following ions plays a key role in generating action potentials in neurons?

(1) Magnesium
(2) Iron
(3) Sodium
(4) Calcium

191. H-zone is present in:

(1) I-band only.
(2) both in A-band and I-band.
(3) light band.
(4) A-band only.

192. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Formation of oxy-haemoglobin	(I)	Due to high pO_2 in the alveoli.
(B)	Fibrosis	(II)	Due to low intra-pulmonary pressure.
(C)	Activation of chemosensitive area	(III)	Due to long exposure to dust.
(D)	Inspiration	(IV)	Due to high CO_2 and H^+ ions.

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-I, D-IV
(2) A-II, B-I, C-III, D-IV
(3) A-II, B-I, C-IV, D-III
(4) A-I, B-III, C-IV, D-II

193. Which among the following belongs to the second largest phylum of kingdom Animalia?

(1) *Octopus* (2) *Laccifer*
(3) *Hirudinaria* (4) *Echinus*

194. Which of the following statements are **correct** regarding skeletal muscle?

A. Muscle bundles are held together by collagenous connective tissue layer called fascicle.
B. Sarcoplasmic reticulum of muscle fibre is a store house of calcium ions.
C. Striated appearance of skeletal muscle fibre is due to distribution pattern of actin and myosin proteins.
D. M line is considered as functional unit of contraction called sarcomere.

Choose the *most appropriate* answer from the options given below.

(1) B and C only (3) C and D only
(2) A, C and D only (4) A, B and C only

195. Refer to the following statements (**A-D**) w.r.t frogs.

A. Frogs respire on land and in the water by two different methods.
B. They have bulged eyes.
C. Tympanum receives sound signals in frogs.
D. The ovaries are functionally connected with kidneys.

Which of the above statement(s) is/are **incorrect**?

(1) A and C only (2) A, B and D only
(3) D only (4) A, B, C and D

196. Given below are two statements:

Statement I: In RNAi, the genes are silenced using dsRNA.

Statement II: Transgenic mice are used in polio vaccine production.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

197. What is the purpose of a gene gun in genetic engineering?

(1) To introduce foreign DNA into plant cells.
(2) To cut DNA at specific sequences.
(3) To amplify DNA segments.
(4) To visualize gene expression.

198. Cortical portion of kidney projecting between the medullary pyramids in human kidney is called:

(1) *vasa recta*.
(2) juxtaglomerular apparatus.
(3) hilum.
(4) column of Bertini.

199. Read the following statements:

A. Large differences arising suddenly in a population.
B. Mutation causes evolution, not minute variations.
C. Mutations are random and directionless.
D. Single step large variation causes speciation.

All these statements represent mechanism of evolution given by:

(1) Charles Darwin (2) Karl Ernst von Baer
(3) Ernst Heckel (4) Hugo deVries

200. Which one of the following is **incorrect** about notochord?

(1) It is present only in larval tail in ascidian.
(2) It is replaced by a vertebral column in adult frogs.
(3) It is absent throughout life in humans from the very beginning.
(4) It is present throughout life in *Amphioxus*.

■■■